Lecture 4

Tuesday, 25 August 2026 7:24 PM

① Example of condn. prob.

② Independence

③ Counting & Sampling.

Disease $P(D) = .01$.

Test T $P(+|D) = .99$ ✓

$P(-|D^c) = .95$ ✓

$P(+ \text{ve test}) = P(+ \cap D) + P(+ \cap D^c)$

$= P(D) \cdot P(+|D) + P(D^c) \cdot P(+|D^c)$ ✓

$= .01 \times .99 + .99 \times .05$

Test +ve. $P(D|+)$

$P(D|+) = \dfrac{P(+|D) \cdot P(D)}{P(+)}$ ($P(D) = .01$)

$= \dfrac{.99 \times .01}{.01 \times .99 + .99 \times .05} \approx .17$ ✓

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$P(D|-)$

$$P(D) = P(D \cap +) + P(D \cap -)$$
$$= P(+) \cdot P(D|+) + P(-) \cdot P(D|-)$$

Independence of Events

$A, B \in F$ are independent if

$$P(A \cap B) = P(A) \cdot P(B)$$

If $P(B) > 0$, same as $P(A) = \frac{P(A \cap B)}{P(B)}$

$$P(A) = P(A|B)$$

$P(A|B)$

E-Commerce App, Subscription Model

1000 users

	Premium (B) subscription	No Premium (B^c)	
Use App (A)	240	360	600
Don't use app (A^c)	160	240	400
	400	600	

$\frac{240}{600} = .4$

$P(B|A) \stackrel{?}{=} P(B)$

$$P(A) = \frac{600}{1000} = .6, \quad P(B) = \frac{400}{1000} = .4$$

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10w

Are A & B independent? $P(A \cap B) = \frac{240}{1000} = .24$

1000

$P(A) \cdot P(B) = .24$

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n events $A_1, A_2 \ldots A_n$ are indep. if for all $I_0 \subseteq \{1, 2 \ldots n\}$,

$$P\left(\bigcap_{i \in I_0} A_i\right) = \prod_{i \in I_0} P(A_i)$$

$P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

Conditional Independence

A, B are Cond. ind. given C if

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C)$$

Counting & Sampling

Discrete Prob. space Ω

$P(\{i\}) = \frac{1}{|\Omega|}$

All outcomes are equally likely



$P(A) = \frac{|A|}{|\Omega|}$

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Basic Counting Rules

① $(C_1, C_2 \ldots C_n)$ → Total no of config is

$m_1 \times m_2 \times m_3 \ldots \times m_n$

$C_1 \to m_1$, $C_2 \to m_2$, $C_n \to m_n$

Eg → (S, P, T) → $4 \times 5 \times 3 = 60$

S → 4, P → 5, T → 3

Eg → (Batch size, Learning Rate, Optimizer)

Batch size → (16, 32, 64) → 3

Learning Rate → (.001, .01) → 2

Optimizer → (Adam, SGD) → 2

$3 \times 2 \times 2 = 12$

Sampling

Basket has n distinguishable balls.

Pick r balls from the basket

① Ordering

(Blue, Red) ≠ (Red, Blue)

② Replacement

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① With ordering & With Replacement

m balls, n draws.

B, R → 2 draws
(B,B), (B,R), (R,B), (R,R)

$(C_1, C_2 \ldots C_n) \longrightarrow m \times m \times \ldots m$

$= m^n$ (4)

C_1: m, C_2: m, C_n: m

Eg → Spotify playlist → 3 songs (S_1, S_2, S_3)

Choose a song. Hit Shuffle. Get another song.

$(S_1, S_1), (S_1, S_2), (S_1, S_3), (S_3, S_1) \rightarrow 3 \times 3 = 9$

② With ordering, without Replacement

$(C_1, C_2 \ldots C_n)$ $\quad m \geq n$

possible outcomes $\longrightarrow m \times (m-1) \times (m-2)$

$= \frac{!m}{!(m-n)} \rightarrow \frac{!3}{!1}$ $\quad \ldots \times (m-(n-1))$ ✓

Eg → Rec 3 cand. (A, B, C)

$m = 3$, $n = 2$

$(\bar{C}_1, \bar{C}_2)$

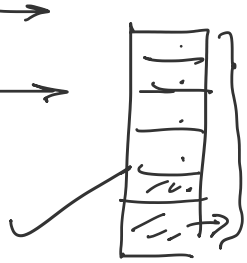
(C, B), (C, A), (B, A), (A, B), (A, C), (B, C)

Without Replacement & Without ordering

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m balls, n draws

$$\binom{m}{n} \rightarrow \frac{!m}{!n\,!(m-n)}$$



Eg - 2 positions
4 candidates
(A, B, C, D)

(A,B), (B,C), (C,D)
(A,C), (B,D), (A,D)

$$\rightarrow \binom{4}{2} = \frac{!4}{!2 \cdot !2} = 6$$

With Replacement & Without ordering

m balls, n draws.

$$\# \{n_1', n_2', n_3 \dots n_m'\}$$

$$n_i \geq 0, \quad \sum_{i=1}^{m} n_i = n$$

possible conf $\rightarrow$

$$\binom{n+m-1}{m-1} = \frac{!(n+m-1)}{!(m-1)\,!n}$$

Eg $\rightarrow$ Popcorn, 2 add-ons, {Ch, Ca, S}

{S, Ca}, {Ch, Ch}, {Ch, Ca}, {Ch, Sa}, {Sa, Sa}, {Ca, Ca}

$n = 2, \; m = 3 \qquad n+m-1 = 4$

$m-1 = 2$

$$\binom{4}{2} = 6$$

{Ch, Sa}
(Sa, Ch) X

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